

Sungil Kim

Department of Industrial Engineering
Ulsan National Institute of Science and Technology
Ulsan, Republic of Korea, 44919

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RESEARCH INTEREST

Quality engineering and management, Time series data representation, Statistical process monitoring and anomaly detection, Industrial statistics and data analytics, AI+manufacturing/logistics

ACADEMIC APPOINTMENT

Ulsan National Institute of Science and Technology , Ulsan, Korea	
Head, Department of Industrial Engineering	September 2024-present
Associate Professor, Department of Industrial Engineering	September 2020-present
Adjunct Professor, Artificial Intelligence Graduate School	September 2020-present
Principal Investigator, Data Analytics Lab	July 2016-present
Director, Industry Intelligitization Institute	February 2025-present
Director, Center for Maritime Data Science	February 2019-April 2021
Assistant Professor, School of Management Engineering	July 2016-August 2020
Director, Center for Advanced Analytics	March 2018-February 2019

EDUCATION

Georgia Institute of Technology , Atlanta, Georgia, USA	
H. Milton Stewart School of Industrial and Systems Engineering	
Ph.D. in Industrial Engineering (with specialization in Statistics)	December 2011
Minor: Supply Chain Management	
M.S. in Statistics	December 2007
M.S. in Industrial Engineering	May 2007
Yonsei University , Seoul, Korea	
Department of Computer Science and Industrial Systems Engineering	
B.S. in Industrial Engineering, Magna Cum Laude	February 2005
Hansung Science High School , Seoul, Korea	February 1998

EMPLOYMENT

Associate Professor, UNIST	July 2016-present
Senior Engineer, Samsung SDS	January 2014-June 2016
Consultant, Terra Technology	September 2011-December 2013
Data Scientist Intern, Predictix	May 2009-December 2009, May 2008-August 2008
Research/Teaching Assistant, Georgia Institute of Technology	May 2006-August 2011

PUBLICATIONS

JOURNALS PAPERS or REFEREED CONFERENCES (Under Review)

1. YongKyung Oh, Giheon Koh, Kyubo Shin, Jae Hoon Moon, and Sungil Kim**, Real-time thyroid disease monitoring with wearable heart rate sensors: a deep learning approach, under review.

JOURNALS PAPERS (Published)

1. YongKyung Oh, JiIn Kwak, and Sungil Kim** (2026), Predicting non-recurrent congestion impact: A pattern-based approach for speed drop ratio prediction using weighted K-nearest Neighbors, *Computers & Industrial Engineering*, 123, pp 111769.
2. YongKyung Oh, Heeyoung Kim, and Sungil Kim** (2025), TSSI: Time Series as Screenshot Images for multivariate time series classification using convolutional neural networks, *Computers & Industrial Engineering*, 209, pp 111393.
3. YongKyung Oh, Giheon Koh, JiIn Kwak, Kyubo Shin, Gi-Soo Kim, Min Joung Lee, Hokyung Choung, Namju Kim, Jae Hoon Moon, and Sungil Kim** (2025), TAOD-Net: a data-driven approach for detecting Thyroid-Associated Orbitopathy symptoms using facial images, *Computers & Industrial Engineering*, 203, pp 111024.
4. YongKyung Oh, Kwonin Yoon, Jaemin Park, and Sungil Kim** (2024), Comparative evaluation of VAE-based monitoring statistics for real-time anomaly detection in AIS data, *Maritime Policy & Management*, 52(4), pp 609-626.
5. YongKyung Oh and Sungil Kim** (2024), Multi-modal lifelog data fusion for improved human activity recognition: a hybrid approach, *Information Fusion*, 110, pp 102464.
6. Jonghwan Mun, Jitae Yoo, Heesun Kim, Nayi Ryu, and Sungil Kim** (2024), Domain knowledge-informed functional outlier detection in the refrigerator inspection lanes, *Computers & Industrial Engineering*, 189, pp 109936.
7. YongKyung Oh, Chiehyeon Lim, Junghye Lee, Sewon Kim, and Sungil Kim** (2024), Multi-channel convolution neural network for gas mixture classification, *Annals of Operations Research*, 339(1), pp 261-295.
8. YongKyung Oh and Sungil Kim** (2023), Grid-Based Bayesian Bootstrap Approach for Real-Time Detection of Abnormal Vessel Behaviors from AIS Data in Maritime Logistics, *IEEE Transactions on Automation Science and Engineering*, accepted.
9. Junsob Jung, Sungil Kim*, and Heeyoung Kim (2023), Spatially-correlated time series clustering using location-dependent Dirichlet process mixture model, *Statistical Analysis and Data Mining*, accepted.
10. YongKyung Oh, Juhui Lee, and Sungil Kim** (2023), Sensor drift compensation for gas mixture classification in batch experiments, *Quality and Reliability Engineering International*, 39(6), pp 2422-2437.
11. Yeram Kim, Chiehyeon Lim*, Junghye Lee, Sungil Kim, Sewon Kim, and Dong-Hwa Seo (2023), Chemistry-informed machine learning: Using chemical property features to improve gas classification performance, *Chemometrics and Intelligent Laboratory Systems*, 237, 104808.

12. YongKyung Oh, JiIn Kwak, JuYeong Lee, and Sungil Kim** (2023), Time delay estimation of traffic congestion propagation due to accidents based on statistical causality, *Electronic Research Archive*, 31(2), pp 691-707.
13. JuYeong Lee, JiIn Kwak, YongKyung Oh, and Sungil Kim** (2023), Quantifying incident impacts and identifying influential factors on urban traffic networks, *Transportmetrica B: Transport Dynamics*, 11(1), pp 279-300.
14. Sungil Kim* (2021), Maximum feasibility estimation, *Information Sciences*, 575, pp 739-801.
15. YongKyung Oh, Hanjoo Kim, Deokjung Lee, and Sungil Kim** (2021), Simulation-based anomaly detection in nuclear reactors, *Journal of the Korean Institute of Industrial Engineers*, 47(2), pp 130-143.
16. YongKyung Oh, Namu Kim, and Sungil Kim** (2021), Transfer Learning based approach for mixture gas classification, *Journal of the Korean Institute of Industrial Engineers*, 47(2), pp 144-159.
17. Sungil Kim, Rong Duan, Guang-Qin Ma, and Heeyoung Kim* (2020), Multiresolution spatial generalized linear mixed model for integrating multi-fidelity spatial count data without common identifiers between data sources, *Spatial Statistics*, 39, 100467.
18. Jaemin Park and Sungil Kim** (2020), Maritime anomaly detection based on VAE-CUSUM monitoring system, *Journal of the Korean Institute of Industrial Engineers*, 46(4), pp 432-442.
19. Sungil Kim* (2019), Revealing household characteristics using connected home products, *Information Sciences*, 486, pp 52-61.
20. Sungil Kim, Heeyoung Kim*, and Jye-Chyi Lu (2019), A practical approach to measuring the impacts of stockouts on demand, *Journal of Business and Industrial Marketing*, 34(4), pp 891-901.
21. Heeyoung Kim, Rong Duan*, Sungil Kim, Jaehwan Lee, and Guang-Qin Ma (2019), Spatial cluster detection in mobility networks: a copula approach, *Journal of the Royal Statistical Society: Series C*, 68(1), pp 99-120.
22. Heeyoung Kim, Justin T. Vastola, Sungil Kim, Jye-Chyi Lu*, and Martha A. Grover (2017), Incorporation of engineering knowledge into the modeling process: a local approach, *International Journal of Production Research*, 55(20), pp 5865-5880.
23. Heeyoung Kim, Sungil Kim, Jian Deng, Jye-Chyi Lu*, K. Wang, C. Zhang, Martha A. Grover, and B. Wang (2017), An integrated holistic model of a complex process, *International Journal of Advanced Manufacturing Technology*, 89(1), pp 1137-1147.
24. Sungil Kim, Heeyoung Kim*, and Yongro Park (2017), Early detection of vessel delays using combined historical and real-time information, *Journal of the Operational Research Society*, 68(2), pp 182-191.
25. Heeyoung Kim, Justin T. Vastola, Sungil Kim*, Jye-Chyi Lu, and Martha A. Grover (2017), Batch sequential minimum energy design with design region adaptation, *Journal of Quality Technology*, 49(1), pp 11-26.
26. Sungil Kim, Heeyoung Kim*, and Younghwan Namkoong (2016), Ordinal classification of imbalanced data with application in emergency and disaster information services, *IEEE Intelligent Systems*, 31(5), pp 50-56.

27. Sungil Kim and Heeyoung Kim* (2016), A new metric of absolute percentage error for intermittent demand forecasts, *International Journal of Forecasting*, 32(3), pp 669-679.
28. Sungil Kim, Heeyoung Kim*, Richard W. Lu, Jye-Chyi Lu, Michael J. Casciato, and Martha A. Grover (2015), Adaptive combined space-filling and D-optimal designs, *International Journal of Production Research*, 53(17), pp 5354-5368.
29. Sungil Kim, Heeyoung Kim*, Jye-Chyi Lu, Michael J. Casciato, Martha A. Grover, Dennis W. Hess, Richard W. Lu, and Xin Wang (2015), Layers of experiments with adaptive combined design, *Naval Research Logistics*, 62(2), pp 127-142.
30. Seung-Kweon Hong*, Sungil Kim (2005), A time prediction model of cursor movement with path constraints, *Journal of the Korean Institute of Industrial Engineers*, 31(4), pp 334-340.

REFEREED CONFERENCES (Published)

1. Jonghun Lee, YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2026), Continuum Dropout for Neural Differential Equations. *Proceedings of the AAAI conference on artificial intelligence. Vol.40..*
2. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2026), FlowPath: Learning Data-Driven Manifolds with Invertible Flows for Robust Irregularly-sampled Time Series Classification. *Proceedings of the AAAI conference on artificial intelligence. Vol.40..*
3. YongKyung Oh, Dong-Young Lim, Sungil Kim**, and Alex A. T. Bui (2025), TANDEM: Temporal Attention-guided Neural Differential Equations for Missingness in Time Series Classification. *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM)*.
4. YongKyung Oh, Keungsu Kam, Jonghun Lee, Dong-Young Lim, and Sungil Kim** (2025), Comprehensive Review of Neural Differential Equations for Time Series Analysis. *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI-25)*.
5. YongKyung Oh, Seungsu Kam, Dong-Young Lim, and Sungil Kim** (2025), Modeling Irregular Astronomical Time Series with Neural Stochastic Delay Differential Equations. **Short Paper**, *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM)*.
6. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2025), Neural Differential Equations for Continuous-Time Analysis. **Tutorial**, *Proceedings of the 34th ACM International Conference on Information and Knowledge Management(CIKM)*.
7. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2025), DualDynamics: Synergizing Implicit and Explicit Methods for Robust Irregular Time Series Analysis. *Proceedings of the AAAI conference on artificial intelligence. Vol.39..*
8. YongKyung Oh, Dong-Young Lim, Sungil Kim, and Alex Bui (2024), Attention-guided Neural Differential Equations Framework for Missingness in Time series Classification. *The 10th Mining and Learning from Time Series Workshop: From Classical Methods to LLMs (KDD MILETS Workshop 2024)*, accepted.
9. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2024), Neural Langevin-type Stochastic Differential Equations for Astronomical time series Classification under Irregular Observations. *In ICLR 2024 Workshop on AI4DifferentialEquations In Science*, accepted.

10. YongKyung Oh, Dong-Young Lim, and Sungil Kim** (2024), Stable neural stochastic differential equations for irregular time series classification, *ICLR (spotlight)*, accepted.
11. YongKyung Oh and Sungil Kim** (2021), Multi-channel convolution neural network for gas mixture classification, *2021 International Conference on Data Mining Workshops (ICDMW)*, pp. 1094-1095.
12. YongKyung Oh and Sungil Kim** (2021), Multichannel convolution neural network for gas mixture classification, *The 4th Conference of Korean Artificial Intelligence Association, Online*.
13. YongKyung Oh and Sungil Kim** (2019), Exploiting Logistics Anomaly Detection using Maritime Big Data. In *IIE Annual Conference. Proceedings*, pp. 964-969, Institute of Industrial and Systems Engineers (IISE).
14. Juhui Lee and Sungil Kim** (2019), Ordinal-imbalanced Data Classification through Data Reduction by Singular Value Decomposing Truncation. In *IIE Annual Conference. Proceedings*, pp. 236-256, Institute of Industrial and Systems Engineers (IISE) [**2019 IIE Best Paper Award, Quality Control & Reliability Engineering Division**].
15. Michael J. Casciato, Sungil Kim, Jye-Chyi Lu, Dennis W. Hess, Martha A. Grover (2012), Optimization of carbon dioxide-assisted nanoparticle deposition process with uncertain design space, *Proceedings of the 11th International Symposium on Process Systems Engineering*, pp 1191-1195.

OTHER PUBLICATIONS AND CREATIVE PRODUCTS

1. YongKyung Oh, Sungil Kim, and Alex AT Bui (2024, August), Deep Interaction Feature Fusion for Robust Human Activity Recognition. In *International Joint Conference on Artificial Intelligence* (pp. 99-116). Singapore: Springer Nature Singapore.
2. YongKyung Oh and Sungil Kim** (2018), Research trends for solving the odor problem, *Industrial Engineering Magazine*, 25(3), 45-51.

INDIVIDUAL STUDENT GUIDANCE

1. Hyejin Cho, MS. Started in Spring 2023; passed defense on December 17, 2025; thesis title: “Assessing User Retention of Updatable Home Appliances: Survival Analysis;” graduated in February 2026; first position:
2. Seongjin Kim, MS. Started in Spring 2024; passed defense on December 15, 2025; thesis title: “A Retrieval-Augmented Generation Framework for Root Cause Analysis in Maritime Accident Adjudication Reports;” graduated in February 2026; first position: BNK Kyongnam Bank
3. Taewon Kang, MS. Started in Fall 2023; passed defense on June 9, 2025; thesis title: “An Unsupervised Framework for Online Failure Prediction and Fault Diagnosis in Industrial Systems;” graduated in August 2025; first position: Posco Future M
4. Jae Hun Cho, MS. Started in Spring 2023; passed defense on December 17, 2024; thesis title: “Title: An Iterative Framework for Active Domain Adaptation in Heterogeneous Image Streams;” graduated in August 2025; first position:

5. Byungkook Koo, MS. Started in Spring 2022; passed defense on June 11, 2024; thesis title: “Enhancing Training Data Quality in Real-Time Streams: An Iterative Framework for Adaptive Domain Adaptation;” graduated in August 2024; first position:
6. Heesun Kim, MS. Started in Spring 2022; passed defense on December 4, 2023; thesis title: “Improving Quality Inspection Performance in Refrigerator Manufacturing Systems through Sensor Calibration and Functional Outlier Detection;” graduated in February 2024; first position: LG Electronics
7. Jitae Yoo, MS. Started in Fall 2021; passed defense on June 4, 2023; thesis title: “Enhanced Deep Anomaly Detection In Contaminated Datasets Using Semi-Supervised Learning;” graduated in August 2023; first position: LX Hausys
8. JiIn Kwak, MS. Started in Spring 2021; passed defense on June 4, 2023; thesis title: “Graph Neural Networks for Traffic Incident Congestion Detection and Impact Prediction;” graduated in August 2023; first position: LG Electronics
9. YongKyung Oh, PhD. Started in Spring 2018; passed defense on December 9, 2022; thesis title: “A Comprehensive Study of Deep Learning for Real-World Multivariate Time Series Classification;” graduated in February 2023; first position: Postdoc at UCLA
10. Jonghwan Mun, MS. Started in Spring 2021; passed defense on December 7, 2022; thesis title: “Domain Knowledge-Informed Functional Outlier Detection for Line Quality Control Systems;” graduated in February 2023; first position: LG Electronics
11. Gyeongjun Kim, MS. Started in Spring 2021; passed defense on December 7, 2022; thesis title: “Semi-Supervised Contrastive Learning for Anomaly Detection in Contaminated Data;” graduated in February 2023; first position: CJ Logistics
12. Giheon Koh, MS. Started in Spring 2021; passed defense on November 29, 2022; thesis title: “Real time Risk Assessment based on Irregularly Sampled Time Series Data from Wearable Devices;” graduated in February 2023; first position: LG Display
13. Kwonin Yoon, MS. Started in Fall 2020; passed defense on June 15, 2022; thesis title: “Vessel Behavior Assessment and Anomaly Detection from AIS Data;” graduated in August 2022; first position: LG Electronics
14. JuYeong Lee, MS. Started in Fall 2019; passed defense on June 17, 2021; thesis title: “Modelling Traffic Incidents Impacts and Prediction of Incident driven Congestion Propagation in a Large Scale Road Network;” graduated in August 2021; first position: Researcher at QRAFT
15. Jaemin Park, MS. Started in Spring 2018; passed defense on December 16, 2020; thesis title: “Unsupervised Anomaly Detection to Monitor Delays of Vessel Voyages based on Variational Autoencoder and Control Chart;” graduated in February 2021; first position: CEO at THY-ROSCOPE Inc.
16. Juhui Lee, MS. Started in Fall 2018; passed defense on June 18, 2020; thesis title: “Data Quality Improvements for Multiclass Classification;” graduated in August 2020; first position: Senior Researcher at THYROSCOPE Inc.

PATENTS

1. Kim, Sungil (primary inventor), Method, system, and operating method for measuring customer retention rate based on survival analysis using log data. (KR 10-2025-0024769, applied February 26, 2025).
2. Kim, Sungil (primary inventor), Apparatus and method for human activity recognition through hybrid fusion of dynamic and static data. (KR 10-2024-0033382, applied March 8, 2024).
3. Kim, Sungil (primary inventor), Method, computer device, and computer program to predict road congestion propagation using pattern matching. (10-2608343, granted November 27, 2023)(KR 10-2021-0120229, applied September 9, 2021)(JP 2022-137508, applied August 31, 2022).
4. Kim, Sungil (primary inventor), Method, computer device, and computer program to predict propagation time delay lag of road congestion using transfer entropy. (10-2604575, granted November 26, 2023)(KR 10-2021-0119772, applied September 8, 2021)(JP 2022-137509, applied August 31, 2022).
5. Kim, Sungil (primary inventor), Method of anomaly detection of vessels applying Bayesian bootstrap. (10-2534357, granted May 16, 2023)(KR 10-2020-0178651, applied December 18, 2020).
6. Kim, Sungil (primary inventor), Encoding apparatus and method for converting time series data into images. (KR 10-2022-0183341, applied December 22, 2022).
7. Kim, Sungil (primary inventor), Sensor drift compensation method and device. (US 18/012,626, applied December 22, 2022)(10-2364019, granted February 14, 2022)(PCT/KR2021/010729, applied August 12, 2021)(KR 10-2020-0101037, applied August 14, 2020).
8. Kim, Sungil (co-inventor), Apparatus for generating target gas detection model based on CNN learning and method thereof. (KR 10-2021-0135617, applied October 13, 2021).
9. Kim, Sungil (primary inventor), Method and apparatus for determining delay possibility of shipment. (10-2250354, granted May 4, 2021), (PCT/KR2020/015854, applied November 12, 2020).
10. Kim, Sungil (primary inventor), Apparatus and method for sensor based realtime odor classification. (10-2106561, granted April 24, 2020), (China 201910234530.5, applied March 26, 2019).
11. Kim, Sungil (primary inventor), Method for risk scoring in logistics system. (KR 10-2015-0150296, applied October 28, 2015).
12. Kim, Sungil (primary inventor), System and method for grid-based geofencing service, (KR 10-2015-0090396, applied June 25, 2015).
13. Kim, Sungil (co-inventor), System and method for detecting and predicting anomalies based on analysis of text data, (KR 10-2014-0142784, applied October 21, 2014).
14. Kim, Sungil (co-inventor), System and method for detecting and predicting anomalies based on analysis of time-series data. (USA 09952921, granted April 24, 2018), (KR 10-2014-0136765, applied October 10, 2014).

15. Kim, Sungil (primary inventor), Apparatus and method for calculating standard route of moving body. (10-1766640, granted August 3, 2017), (KR 10-2014-0127859, applied September 24, 2014).
16. Kim, Sungil (primary inventor), Apparatus and method for early detection of abnormality. (KR 10-2014-0112321, applied August 27, 2014).

GRANTS & CONTRACTS

1. The AI Era, A Knowledge Platform for All: Libraries as a Social Operating System for Sustainable Growth
 - Role: Group Leader
 - Source: Global Humanities and Social Sciences Convergence Research Support Project, National Research Foundation of Korea (NRF)
 - Duration: September 1, 2025-August 31, 2030
 - Amount: ₩10,000,000,000
2. Ulsan Industry-University Convergence Zone: Enterprise-Focused Industry-University Convergence Promotion Support
 - Role: Principal Investigator
 - Source: Industry-University Convergence District Creation Project, Korea Institute for Advancement of Technology (KIAT)
 - Duration: January 1, 2025-December 31, 2027
 - Amount: ₩735,000,000
3. Development of Domain-free Time Series Data Drift Management Techniques
 - Role: Principal Investigator
 - Source: Mid-Career Researcher Program, National Research Foundation of Korea (NRF)
 - Duration: March 1, 2025-February 29, 2028
 - Amount: ₩650,000,000
4. Development of Foundation Models for Bioelectrical Signal Data and Validation of Their Clinical Applications: A Noise-and-Variability Robust, Generalizable Self-Supervised Learning Approach
 - Role: Participant
 - Source: IITP
 - Duration: July 1, 2024-December 31, 2026
 - Amount: ₩2,760,000,000
5. Human Centered-Carbon Neutral Global Supply Chain Research Center
 - Role: Group Leader (Data Science Group)
 - Source: Engineering Research Center (ERC), National Research Foundation of Korea (NRF)

- Duration: June 1, 2023-May 31, 2030
 - Amount: ₩13,500,000,000
6. Enhancement of Fault Detection/Discrimination Models in LQC Processes
 - Role: Principal Investigator
 - Source: LG Electronics
 - Duration: June 30, 2023-June 28, 2024
 - Amount: ₩50,000,000
 7. Developing data-driven user satisfaction metrics for ThinQ contents in LG Upgradable home appliances
 - Role: Principal Investigator
 - Source: LG Electronics
 - Duration: June 1, 2023-May 31, 2024
 - Amount: ₩80,000,000
 8. Development of a robust methodology for thyroid eye disease recognition based on Heterogeneous devices
 - Role: Participant
 - Source: UNIST-THYROSCOPE Inc.
 - Duration: March 1, 2023-December 31, 2024
 - Amount: ₩200,000,000
 9. Thyroid eye disease recognition based on Heterogeneous cameras
 - Role: Participant
 - Source: Korea Innovation Foundation
 - Duration: October 1, 2022-January 31, 2023
 - Amount: ₩60,000,000
 10. Research on AI solutions for predicting appropriate thyroid drug dosage
 - Role: Principal Investigator
 - Source: TIPA(the Korea Technology and Information Promotion Agency for SMEs)
 - Duration: May 1, 2022-April 30, 2024
 - Amount: ₩75,864,000
 11. Damage detection of linear compressors for refrigerators in line quality control system
 - Role: Principal Investigator
 - Source: LG Electronics
 - Duration: November 29, 2021-November 28, 2022
 - Amount: ₩50,000,000
 12. AI Innovation Hub R&D

- Role: Participant
 - Source: IITP
 - Duration: July 1, 2021-December 31, 2025
 - Amount: ₩1,837,500,000
13. Thyroid eye disease recognition using convolutional neural networks
- Role: Participant
 - Source: Korea Innovation Foundation
 - Duration: August 1, 2021-December 31, 2021
 - Amount: ₩61,000,000
14. Development of biosignal error correction model between wearable devices
- Role: Principal Investigator
 - Source: Ulsan Industry University Convergence Institute
 - Duration: September 1, 2021-December 31, 2021
 - Amount: ₩19,400,000
15. Development of personalized self-management solutions and disease risk prediction techniques for thyroid disease management
- Role: Participant
 - Source: UNIST AI Innovation Park
 - Duration: August 1, 2021-July 31, 2022
 - Amount: ₩100,000,000
16. Development of AI-based vessel ETA (Estimated Time of Arrival) prediction methodology for intelligent smart ports
- Role: Principal Investigator
 - Source: National Research Foundation of Korea (NRF)
 - Duration: June 1, 2021-February 29, 2024
 - Amount: ₩131,343,000
17. Development of AI technology for identifying traffic congestion patterns
- Role: Participant
 - Source: NAVER
 - Duration: November 1, 2020-October 31, 2022
 - Amount: ₩300,000,000
18. AI dataset construction for manufacturing facilities
- Role: Participant
 - Source: the Korea Technology and Information Promotion Agency for SMEs (TIPA)
 - Duration: Sep 18, 2020-November 30, 2020

- Amount: ₩283,739,091
19. Manufacturing data analysis and AI model development
 - Role: Participant
 - Source: interX
 - Duration: June 10, 2020-November 30, 2020
 - Amount: ₩122,727,273
 20. Incorporation of domain knowledge into the modeling process for quality improvement in smart manufacturing
 - Role: Principal Investigator
 - Source: National Research Foundation of Korea (NRF)
 - Duration: March 1, 2017-February 28, 2022
 - Amount: ₩250,000,000
 21. Structural analysis on the processes of technological innovation in the fourth industrial revolution: focusing on open innovation and concurrent transformation of regional innovation system
 - Role: Participant
 - Source: National Research Foundation of Korea (NRF)
 - Duration: July 1, 2018-June 30, 2021
 - Amount: ₩228,762,000
 22. Development of AI-Based Reactor Core Diagnostics System
 - Role: Participant
 - Source: Korea Hydro & Nuclear Power Co.,Ltd.
 - Duration: June 1, 2019-May 31, 2021
 - Amount: ₩446,557,680
 23. Development of stowage optimization engine solver using reinforcement learning
 - Role: Principal Investigator
 - Source: Cyberlogitec
 - Duration: November 1, 2018-August 31, 2020
 - Amount: ₩45,000,000
 24. Development of Algorithms for Mixture Gases Classification
 - Role: Principal Investigator
 - Source: Ulsan Industry-University Convergence Campus
 - Duration: March 16, 2020-July 31, 2020
 - Amount: ₩40,000,000
 25. Smart Port Logistics Support Center

- Role: Principal Investigator
 - Source: Ulsan Port Authority
 - Duration: January 1, 2019-December 31, 2019
 - Amount: ₩1,395,460,000
26. A Constraint satisfaction problem with attribute data: an application to connected home products
- Role: Principal Investigator
 - Source: UNIST
 - Duration: September 1, 2016-August 31, 2019
 - Amount: ₩20,000,000
27. Data mining project lab
- Role: Participant
 - Source: UNIST-Taesung Environmental Research Institute
 - Duration: October 1, 2018-December 31, 2018
 - Amount: ₩15,000,000
28. Development of data analytics methods for mixture gases classification
- Role: Principal Investigator
 - Source: UNIST-Taesung Environmental Research Institute
 - Duration: August 1, 2018-December 31, 2018
 - Amount: ₩10,000,000
29. Blockchain-based system engineering
- Role: Participant
 - Source: UNIST
 - Duration: March 1, 2018-December 31, 2018
 - Amount: ₩45,000,000
30. Development of data analytics methods to identify the sources of odor
- Role: Principal Investigator
 - Source: Ulsan Industry-University Convergence Campus
 - Duration: October 1, 2017-March 31, 2018
 - Amount: ₩38,500,000
31. Development and application of methods and an intelligent platform system for Industry 4.0
- Role: Participant
 - Source: UNIST
 - Duration: February 1, 2017-December 31, 2017
 - Amount: ₩40,000,000

TEACHING EXPERIENCE

Ulsan National Institute of Science and Technology, Ulsan, Korea

Department of Industrial Engineering

Instructor

- IE 362/MGE 362: Statistical Quality Management Spring, 2017-2025
- IE552/AI603: Neural Differential Equations Fall, 2024
- IE 313: Time Series Analysis Fall, 2023
- IE 509/AI 533: Advanced Quality Control Fall, 2021-2022
- AI 590: AI Graduate Seminar Spring, 2022
- AI 501: Introduction to AI Spring, 2022
- IE 471: Special Topic (Project Lab) Fall, 2021
- MGT 101: Entrepreneurship & Big Data Fall, 2020
- IE 502/MGE 502: Statistical Programming Fall, 2018-2020
- TIM 713: Industrial Innovation Seminar Fall, 2018
- MGE 509: Advanced Quality Control Fall, 2017-2019
- MGE 301: Operations Research I Fall, 2016-2017
- MGT 209: Operations Management Fall, 2016

Georgia Institute of Technology, Atlanta, Georgia, USA

H. Milton Stewart School of Industrial and Systems Engineering

Teaching Assistant

- ISyE 4031: Regression and Forecasting Spring, 2011
- ISyE 3770: Probability and Statistics Fall, 2010
- ISyE 2027: Probability with Applications Spring, 2010
- ISyE 4803: Advanced Supply Chain Logistics Fall, 2007
- ISyE 6739: Statistical Methods Summer, 2007

PRESENTATIONS

INTERNATIONAL

1. Nonparametric and Recurrent Survival Models for User Retention , *INFORMS Annual Meeting*, October 2025.
2. Assessing User Retention of Updatable Home Appliances: Survival Analysis, *IISE Annual Conference & Expo 2025*, May 2025.
3. Grid-Based Bayesian Bootstrap Approach for Real-Time Detection of Abnormal Vessel Behaviors from AIS Data in Maritime Logistics, *2024 IEEE 20th International Conference on Automation Science and Engineering*, August 2024.
4. **(Invited Talk)** Domain-Knowledge-Informed Functional Outlier Detection for Line Quality Control Systems, *2nd INFORMS Conference on Quality, Statistics and Reliability*, July 2024.

5. **(Invited Talk)** Integrating Neural Controlled Differential Equations and Neural Flow for Comprehensive Irregularly-sampled Time Series Analysis, *IISE Annual Conference & Expo 2024*, May 2024 [**2024 Best Track Paper Competition Finalist, Data Analytics and Information Systems (DAIS) Division**].
6. Optimal and Personalized Dose Determination for Patients with Thyroid Hormone Disorders Using Deep Learning-Based Survival Analysis, *IISE Annual Conference & Expo 2024*, May 2024.
7. Enhancing Human Activity Recognition with Comparative Data Fusion and Learning Strategies, *IISE Annual Conference & Expo 2024*, May 2024.
8. Enhancing Astronomical time series Classification with Neural Stochastic Differential Equations under Irregular Observations, Knowledge Guided ML (KGML) Bridge Program, *The 38th Annual AAAI Conference on Artificial Intelligence 2024* (Poster), February 2024.
9. Invertible Solution of Neural Differential Equations for Analysis of Irregularly-Sampled Time Series, Artificial Intelligence for Time Series (AI4TS) Workshop, *The 38th Annual AAAI Conference on Artificial Intelligence 2024* (poster), February 2024.
10. Unifying Neural Controlled Differential Equations and Neural Flow for Irregular Time Series Classification, *The Symbiosis of Deep Learning and Differential Equations Workshop, NeurIPS 2023 Workshop* (poster), November 2023.
11. **(Invited Talk)** Comparative evaluation of VAE-based monitoring statistics for real-time anomaly detection in AIS data, The 11th International Conference on Logistics and Maritime Systems (LOGMS 2023), September 2023.
12. Real-time risk assessment of thyroid function abnormality using irregularly-sampled heart rate records, *IISE Annual Conference & Expo 2023*, May 2023.
13. Sparse multi-channel convolutional neural network for multivariate time series classification, *IISE Annual Conference & Expo 2023*, May 2023.
14. TAOD-Net: Near-ophthalmologist level active thyroid-associated orbitopathy detection on frontal eye photographs with deep learning, *IISE Annual Conference & Expo 2023*, May 2023.
15. Domain knowledge-informed functional outlier detection for line quality control systems, *IISE Annual Conference & Expo 2023*, May 2023.
16. **(Invited Talk)** Neural stochastic differential equations based on the Ornstein-Uhlenbeck process, *IISE Annual Conference & Expo 2023*, May 2023 [**2023 Best Track Paper Competition Finalist, Data Analytics and Information Systems (DAIS) Division**].
17. **(Invited Talk)** Domain knowledge-informed functional outlier detection for line quality control systems, *Industrial and Systems Engineering Virtual Seminar*, Rutgers University, March 2023.
18. **(Invited Talk)** Time delay estimation of traffic congestion based on statistical causality, *3rd Workshop on Data-driven Intelligent Transportation (DIT 2022)*, Held in conjunction with CIKM 2022, October 2022.
19. Sensor drift compensation for gas mixture classification in batch experiments, *IISE Annual Conference & Expo*, May 2022.

20. A propagation prediction method for non-recurrent traffic congestion, *IISE Annual Conference & Expo*, May 2022.
21. **(Invited Talk)** Time delay estimation of traffic congestion based on statistical causality, *The 5th International Conference on Econometrics and Statistics*(EcoSta 2022), June 2022.
22. **(Invited Talk)**Detection of abnormal vessel behaviors from AIS data using the Bayesian bootstrap, *Industrial and Systems Engineering Virtual Seminar*, Rutgers University, November 2021.
23. A propagation prediction method for non-recurrent traffic congestion, *INFORMS Annual Meeting*, October 2021.
24. VAE-CUSUM: a new feature-based monitoring chart to monitor vessel voyages using AIS data, *INFORMS Annual Meeting*, October 2021.
25. Multi-channel convolution neural network for gas mixture classification, *IISE Annual Conference & Expo 2021*, May 2021.
26. Logistics anomaly detection with maritime big data: a bootstrap approach, *IISE Annual Conference & Expo 2021*, May 2021.
27. Sensor drift compensation for mixed gas classification under batch experiments, *INFORMS Annual Meeting*, November 2020.
28. **(Invited Talk)** Maximum feasibility estimation, *The 2020 INFORMS Workshop on Data Science*, November 2020.
29. Logistics anomaly detection with maritime big data: a bootstrap approach, *IISE Annual Conference & Expo 2020*, November 2020.
30. Maximum feasibility estimation, *INFORMS Annual Meeting*, October 2019.
31. **(Invited Talk)** Revealing household characteristics using connected home products, *The Fifth International Conference on the Interface between Statistics and Engineering*, June 2019.
32. Ordinal-imbalanced data classification by singular value decomposing truncation, *IISE Annual Conference & Expo 2019*, May 2019.
33. Exploiting logistics anomaly detection using maritime big data, *IISE Annual Conference & Expo 2019*, May 2019.
34. Early detection of vessel delays using combined historical and real-time information, *INFORMS Annual Meeting*, October 2017.
35. Layers of experiments with adaptive combined design, *INFORMS Annual Meeting*, November 2014.
36. **(Invited Talk)** Prediction & inference using the hierarchical spatiotemporal varying coefficient model: applied to detailed sales forecasting for retail providers, *Georgia Tech Research and Innovation Conference*, February 2010.
37. Detailed sales forecasting & promotion analysis for retail providers, *Joint Statistical Meetings*, August 2009.

DOMESTIC

1. Learning from Patient Similarity: A Cluster-Based Approach to ECG Imputation, KIIE, November 2025.
2. Integrating Environmental Data into Decision-Making Based Maritime Anomaly Detection, KIIE, November 2025.
3. Missing Mechanism-Aware Diagnosis for Multivariate Time Series Imputation, November 2025.
4. Domain-Specific Foundation Models for Initial Bio-device Deployment, KIIE, November 2025.
5. Deep Dynamic Survival Modeling for Personalized Risk Prediction under Competing Risks, KIIE, June 2025.
6. Improving Ship Fuel Consumption Prediction via Environmental Data Integration, KIIE, June 2025.
7. Statistical Modeling of Rolling Retention via Recurrent Survival Analysis, KIIE, June 2025.
8. **(Invited Talk)** Stable Neural Stochastic Differential Equations for Irregular Time Series Classification, KDMS Conference, November 2024.
9. Addressing Spatial Gaps in XCO₂ Monitoring: A Machine Learning Approach for South Korea, KIIE, October 2024.
10. Personalized Initial Dosage Recommender System for Hyperthyroidism Patients Using Deep Survival Analysis, KIIE, May 2024.
11. Assessing User Retention of Upgradable Home Appliances: Survival Analysis, KIIE, May 2024.
12. Optimal and personalized dose determination for patients with thyroid hormone disorders using deep learning-based survival analysis, KIIE, November 2023.
13. An intelligent diagnostic system for thyroid-associated ophthalmopathy using images from heterogeneous smartphone devices, KIIE, November 2023.
14. **(Invited Talk)** Heterogeneous model fusion for enhanced sensor-based human activity recognition, Korea Computer Congress (KCC), June 2023 [*2023 KCC Best Presentation Award, AI Division*].
15. **(Invited Talk)** Flow-based neural differential equations for time series analysis, Korea Computer Congress (KCC), June 2023 [*2023 KCC Best Paper Award, AI Division*].
16. Spatio-temporal graph neural network approach for real-time traffic incident detection, KIIE, June 2023.
17. Flow-based neural differential equations for time series analysis, KIIE, June 2023.
18. Heterogeneous model fusion for enhanced sensor-based human activity recognition, KIIE, June 2023.
19. Real-time detection of changing-state active galactic nuclei using a deep learning approach, KIIE, June 2023.
20. **(Invited Talk)** Lifelog data fusion approach for emotion recognition, Conference of Korea Software Congress 2022 (KSC 2022), December 2022.
21. **(Invited Talk)** Irregularly sampled time series classification using neural stochastic differential equation, Conference of Korean Artificial Intelligence Association, November 2022.

22. **(Invited Talk)** Real-time risk assessment of thyroid function abnormality using irregularly-sampled heart rate records, Conference of Korean Artificial Intelligence Association, November 2022.
23. Irregularly sampled time series classification using neural stochastic differential equation, KDMS Conference, November 2022.
24. Irregularly sampled time series classification using neural stochastic differential equation, KIIE, November 2022.
25. Deep learning approach for behavior of piston of a linear compressor, KIIE, November 2022.
26. Domain knowledge-informed functional outlier detection for line quality control systems, KIIE, June 2022.
27. Multivariate time series classification using multi channel CNN classifier, KIIE, June 2022.
28. TAED-Net: Near ophthalmologist-level thyroid-associated eye symptom detection on frontal eye photographs with deep learning, KIIE, June 2022.
29. Risk assessment of abnormal vessel behaviours from AIS data using elastic depths, KIIE, June 2022.
30. **(Invited Talk)** Mixture gas classification and sensor drift compensation, LG H&A-UNIST Future Webinar 21, September 2021.
31. Commercial area analysis using big data on GIS: case of YongIn-si, KIIE, May 2021.
32. Sensor drift compensation for mixed gas classification under batch experiments, KIIE, May 2021.
33. Transfer-learning based approach for mixture gas classification, KIIE, November 2020.
34. Congestion propagation modeling with graph neural network (GNN), KIIE, November 2020.
35. Simulation-based anomaly detection in nuclear reactors, KIIE, November 2020.
36. Exploiting logistics anomaly detection using maritime big data, KIIE, November 2019.
37. Ordinal-imbalanced data classification through noise reduction by singular value decomposing truncation, KIIE, April 2019.
38. **(Invited Talk)** Data analytics in logistics systems: monitoring, classification, and assessment, Yonsei University, May 2018.
39. Sensor drift compensation using temperature and humidity for mixture gases classification, KIIE, April 2018.
40. **(Invited Talk)** Big data analytics in logistics, 4th UNIST Big Data Symposium, November 2017.

SERVICE

Campus Contributions

- Head, Department of Industrial Engineering, 2024-2026
- Member, UNIST Artificial Intelligence Committee, 2025-2026
- Chair, The Industrial Cooperation Working Group, UNIST-LG Electronics, 2023-2026
- Member, UNIST Academic Steering Committee, 2022-2024

- Chair, The Industrial Cooperation Committee, Artificial Intelligence Graduate School, 2021-2024
- Chair, The Faculty Recruitment Committee, Department of Industrial Engineering, 2021-2022
- Member, The Faculty Recruitment Committee, Department of Industrial Engineering, 2023
- Chair, The Department Education Committee, Department of Industrial Engineering, 2020-2023
- Member, UNIST-KAIST-POSTECH Data Science Competition Founding Committee, 2021
- Member, The University Admission Interview Committee, 2020
- Member, Election Commission for Faculty Representatives of the UNIST Council, 2020
- Member, The Fourth Industrial Revolution Working Committee, 2018-2022
- Member, UNIST Visibility Committee, 2018-2020
- Member, Space Planning and Allocation Committee for Industry-University Convergence Campus, 2018-2019
- Member, The Faculty Recruitment Committee, School of Management Engineering, 2017-2020
- Member, The Undergraduate Admissions Committee, 2017
- Member, Committee for the Preparation of the Movement to Complex Campus, School of Management Engineering, 2017

Professional Membership

- Senior Member, The Institute for Operations Research and the Management Sciences (INFORMS), 2024-present
- Member, The Institute for Operations Research and the Management Sciences, 2006-2024
- Member, The Institute of Industrial & Systems Engineers, 2019-present
- Member, The Korean Institute of Industrial Engineers, 2014-present

Public and Community Service

- Member (Overall/Planning), Organizing Committee, the KIIE Annual Conference, Fall 2023
- Director, Korea Data Mining Society (KDMS), 2022-2026
- Member, Logistics Policy Committee, Ulsan Metropolitan City Hall, 2019-2021, 2023-2024

Professional Activities

- Board of Directors, IISE QCRE Division, 2025-2026
- QCRE Track Program Chair, *IISE Annual Conference & Expo 2025*, 2024-2025
- **Area editor in Statistics, Quality, Reliability & Maintenance**, *Computers & Industrial Engineering*, 2023-present
- **Associate editor**, *Journal of the Korean Institute of Industrial Engineers*, 2024-present
- Session chair of the KIIE Annual Conference, ML & AI, Daejeon, November 2025.
- Session chair of the KIIE Annual Conference, Business Analytics, Jeju, June 2022.

- Invited reviewer, The Faculty Recruitment Committee, Department of Industrial & Systems Engineering, KAIST, 2021-2022
- Session chair of the INFORMS Annual Meeting, VTA08. Data Mining II, October 2021.
- Reviewer of the Mid-Career Researcher Program, National Research Foundation of Korea, 2021
- Session chair of the KIIE Annual Conference, Probability/Statistics/Quality, Jeju, June 2021.
- Reviewer, IISE Transactions, IEEE Transactions on Automation Science and Engineering, Annals of Operations Research, 2021-present
- Session chair of the Fifth International Conference on the Interface between Statistics and Engineering (ICISE), Statistics and Analytics, Seoul, June 2019.
- Session chair of the KIIE Annual Conference, Industrial AI, Gwangju, April 2019.

PROFESSIONAL EXPERIENCE

Samsung SDS, Seoul, Korea

Data Analytics Lab, Algorithm Research Team, R&D Center

Senior Engineer/Data Scientist

January 2014-June 2016

- Developed risk assessment and scoring algorithm using text mining for SDS Smart Logistics portal solution, Cello Square.
- Developed a data-driven method for early detection of vessel delays combined with real-time vessel tracking information. The proposed approach is validated by applying real datasets extracted from the logistics platform of Samsung SDS, Cello.
- Analyzed global IT trends and disruptive technologies in the fields of big data analytics, the Internet of Things (IoT), and video analytics.
- Collaborated with SDS R&D center in San Jose to research new technologies in data analytics for SDS mid/long term business in global markets.

Terra Technology, Norwalk, Connecticut, USA

Supply Chain Management Consultant

September 2011-December 2013

- Applied data mining techniques (e.g., pattern recognition, regression analysis, clustering) to real business problems.
- Collected/identified big data from existing internal databases and external/public data sources.
- Cooperated with the client's DBA team and tested the performance of forecasts.
- Developed technical solutions, including writing and testing SQL procedures and/or UNIX/Windows shell scripts to meet customer integration requirements.
- Participated in demand planning projects for Procter & Gamble, Unilever, Kraft Foods Inc., ConAgra Foods, Kellogg's, etc.
- Analyzed data to identify demand volatility and changing consumer preferences.
- Created data validation templates using Qlikview reports by adding new functionalities: multiple selections, high dimensional search.

Predictix, Atlanta, Georgia, USA

Data Scientist Intern

May-December 2009 & May-August 2008

- Participated in pre-sales projects for Target Corporation.
- Developed a synthetic sales data generator for studying promotion effects.
- Analyzed retail sales transaction data for promotion planning.
- Developed spatial-temporal data mining techniques for retail sales transaction data from Shopko.
- Collected/identified big data from existing internal databases and external/public data sources.

Georgia Institute of Technology, Atlanta, Georgia, USA

H. Milton Stewart School of Industrial and Systems Engineering

Research Assistant

May 2006-August 2011

- Developed statistical methodologies for a data collection plan with uncertain design space.
- Developed a new approach, *Layers of Experiments*, for the robust optimization of nanoparticle synthesis.
- Explored a new statistical methodology in a multi-level, multi-scale framework in the context of supply chain logistics systems.
- Performed inventory system analysis on stocked items and participated in a warehousing design for Enraf Fluid Technology.

TECHNICAL SKILLS

R, Python, Julia, MATLAB, Hadoop, Spark, Perl, C/C++, Korn Shell, Unix/Linux, Java, SQL, MS Access, PL/SQL, XML, MySQL, LINDO/LINGO, GAMS, Qlikview

AWARDS & HONORS

- 2023 Excellence in Mentoring Award, *College of Information and Biotechnology, UNIST*
- 2023 KCC Best Presentation Award, AI Division, *Korean Institute of Information Scientists and Engineers*
- 2023 KCC Best Paper Award, AI Division, *Korean Institute of Information Scientists and Engineers*
- 2023 IISE Best Paper Competition Finalist, Data Analytics and Information Systems Division, *The Institute of Industrial and Systems Engineers*
- 2021 IISE Best Paper Award, Logistics and Supply Chain Division, *The Institute of Industrial and Systems Engineers*
- 2019 IISE Best Paper Award, Quality Control & Reliability Engineering Division, *The Institute of Industrial and Systems Engineers*
- 2015 SDSers of the Quarter, *Samsung SDS*
- 2010 Honorable mentions, *The Business and Economics Statistics Section at the Joint Statistical Meetings*
- 2006 Global Logistics Scholar, *The Logistics Institute, Georgia Institute of Technology*

- 2005 Graduate Research Scholarship, *Korea Research Foundation*